

**CO2 ASSESSMENT TOOL 2**  
**SCENARIO-BASED QUESTIONS**

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**Code of Conduct:**

**I K. Hemakshitha(192425088) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment I understand that any violation of academic integrity rules will result in disciplinary action.**

**1. Object Boundary Detection**

**(a) Interpret the scenario and explain the importance of boundary detection.**

In a medical image, different tissues and organs may have different structures. Boundary detection is used to identify the edges or borders between these tissues. It helps to separate one tissue from another and makes the shape and location of organs or abnormal regions clearer.

Boundary detection is important because it can help doctors identify tumors, lesions, organs, blood vessels, and other abnormal regions. It is also useful in medical image segmentation and diagnosis.

**(b) Identify key challenges in detecting boundaries.**

The main challenges are:

1. Noise: Medical images may contain unwanted noise that can create false edges.
2. Low contrast: Two tissues may have very similar intensity values, making their boundary difficult to detect.
3. Blurred boundaries: Some tissue boundaries may not be sharp.
4. Weak edges: Some important boundaries may have only a small change in intensity.

5. Complex structures: Medical images can contain many tissues close to each other.

**(c) Apply a suitable edge detection method.**

A suitable method is the Canny Edge Detection method.

The basic steps are:

1. Convert the image to grayscale if required.
2. Apply Gaussian filtering to reduce noise.
3. Calculate the intensity gradient.
4. Apply non-maximum suppression to make edges thin.
5. Use double thresholding to identify strong and weak edges.
6. Perform edge tracking by hysteresis to obtain connected edges.

**(d) Justify your selection with logical reasoning.**

Canny edge detection is suitable because medical images can contain noise and weak boundaries. The Gaussian filter reduces noise before detecting edges. It can detect both strong and weak edges and produces thin and accurate boundaries.

Compared with a simple edge detector, Canny generally provides better edge localization and reduces false edges.

**(e) Present your explanation clearly and systematically.**

Input: Medical image

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Noise reduction: Gaussian filter

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Gradient calculation

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Non-maximum suppression

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Double thresholding

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Edge tracking

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Output: Clear tissue boundaries

Conclusion: Canny Edge Detection is a good choice for medical images because it reduces noise and detects important tissue boundaries accurately.

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## **2. Simple Segmentation**

### **(a) Interpret the scenario and explain segmentation requirements.**

In this scenario, the object and background can be clearly distinguished based on their intensity values. The main requirement is to divide the image into two regions:

- Object
- Background

Segmentation makes it easier to identify and analyze the required object separately from the background.

### **(b) Identify key features enabling separation.**

The important features are:

1. Different intensity values: The object and background have different brightness levels.
2. Clear separation: There is a noticeable difference between object and background.
3. Simple image structure: The objects do not have complicated overlapping regions.
4. Low noise: The image is relatively clean.

For example, if the object has intensity values greater than 150 and the background has values below 150, a threshold can be selected around 150.

### **(c) Apply a suitable segmentation technique.**

The suitable technique is Thresholding.

The basic process is:

1. Select a threshold value  $T$ .

2. Compare each pixel with T.
3. If the pixel value is greater than T, classify it as the object.
4. Otherwise, classify it as background.

Mathematically:

If  $f(x,y) > T \rightarrow \text{Object}$

If  $f(x,y) \leq T \rightarrow \text{Background}$

**(d) Justify your decision logically.**

Thresholding is suitable because the object and background have clearly different intensity values. It is simple and requires less computation.

It is also easy to implement and works well when the image has good contrast between the object and background.

**(e) Present your response clearly.**

Input image

↓

Check intensity values

↓

Select threshold T

↓

Compare each pixel with T

↓

Object / Background classification

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Segmented image

Conclusion: Thresholding is the best simple method when the object and background have clearly different intensity values. It provides fast and easy segmentation.

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### **3. Complex Segmentation**

**(a) Interpret the scenario and explain segmentation challenges.**

In this scenario, the image contains multiple regions, but their intensity values overlap. This makes it difficult to separate the regions using simple thresholding.

For example, two different objects may have almost the same intensity values. Therefore, intensity alone cannot provide accurate segmentation.

**(b) Identify key issues such as intensity overlap.**

The main issues are:

1. Intensity overlap: Different objects may have similar pixel values.
2. Multiple regions: The image contains several objects.
3. Touching objects: Two objects may be connected.
4. Noise: Noise can affect the segmentation result.
5. Irregular shapes: Objects may have complex shapes.
6. Weak boundaries: Some regions may not have clear boundaries.

Because of these problems, a simple thresholding technique may not be sufficient.

**(c) Apply an appropriate advanced segmentation method.**

A suitable method is Watershed Segmentation.

The basic steps are:

1. Preprocess the image to reduce noise.
2. Convert the image into a suitable grayscale or gradient representation.
3. Identify important regions or markers.
4. Apply the watershed algorithm.
5. Separate connected or overlapping objects.
6. Obtain the final segmented regions.

**(d) Justify how the method handles complexity.**

Watershed segmentation is useful for images containing touching or overlapping objects. It considers the image as a topographic surface with high and low regions.

The algorithm creates boundaries between different regions and can separate objects that are connected to each other.

For example, if two objects touch each other, watershed can identify a boundary between them and separate them into two different regions.

**(e) Provide a structured and clear explanation.**

Input complex image



Noise reduction / preprocessing



Gradient or distance transformation



Identify markers



Apply Watershed algorithm



Separate overlapping regions



Final segmented image

Conclusion: Watershed segmentation is suitable for complex images because it can separate touching and overlapping regions more effectively than simple thresholding.

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## **4. Broken Edges**

(a) Interpret the scenario and explain why edges are broken.

After applying edge detection, object boundaries may appear as broken or discontinuous lines instead of complete boundaries.

This can happen because some parts of the boundary have weak intensity changes. Noise can also cause missing or false edge pixels.

For example, an object boundary may appear as:

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instead of a continuous line:

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**(b) Identify key issues affecting boundary continuity.**

The main issues are:

1. Noise: Noise can remove or disturb edge pixels.
2. Weak edges: Some boundary pixels may have small intensity differences.
3. Low contrast: The object and background may have similar intensity.
4. Missing pixels: Some parts of an edge may not be detected.
5. Small gaps: Edge detection can create small gaps between neighboring edge pixels.

These problems make it difficult to obtain a complete object boundary.

**(c) Apply a suitable method to improve edge connectivity.**

A suitable method is Morphological Closing.

Morphological closing generally uses:

Dilation → Erosion

A suitable structuring element is applied to the detected edge image.

**(d) Justify your approach logically.**

Morphological closing is suitable because it can connect nearby broken edges and fill small gaps.

First, dilation expands the edge pixels and connects nearby gaps. Then erosion removes the extra expansion while maintaining the connected structure.

Therefore, broken boundaries become more continuous.

**(e) Present your explanation clearly.**

Input image

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Edge detection

↓

Broken edges obtained

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Morphological dilation

↓

Connect nearby gaps

↓

Morphological erosion

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Continuous edges

Conclusion: Morphological closing is useful for improving edge connectivity because it fills small gaps and connects nearby broken boundaries.

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## **5. Combined Requirement**

### **(a) Interpret the scenario and define the combined requirement.**

In an industrial inspection system, images are used to detect defects such as cracks, scratches, holes, or damaged areas.

The image may contain noise due to the camera, lighting, or manufacturing environment. At the same time, important defect boundaries must be preserved.

Therefore, the combined requirement is:

Remove unwanted noise + Preserve important edges

### **(b) Identify key issues involving noise and edge preservation.**

The main issues are:

1. Image noise: Noise may create unwanted pixels and false edges.
2. Small defects: Important defects may be small and can easily be removed.
3. Over-smoothing: Strong smoothing may remove useful edges.
4. False edges: Noise may be detected as a defect.
5. Weak defect boundaries: Some defects may have low contrast.

Therefore, the filtering method must remove noise without destroying important information.

### **(c) Apply an appropriate sequence of techniques.**

A suitable sequence is:

Median Filter → Canny Edge Detection

### Step 1: Median Filtering

The median filter is first applied to the image.

Its purpose is to remove noise, especially salt-and-pepper noise, while preserving edges better than a simple averaging filter.

### Step 2: Canny Edge Detection

After noise removal, Canny Edge Detection is applied.

It detects important edges and boundaries of defects.

### **(d) Justify the order and selection with strong reasoning.**

The order is important.

If Canny edge detection is applied directly to a noisy image, the noise may be detected as false edges. This can lead to incorrect defect detection.

Therefore, we first use the Median Filter to reduce noise.

After noise reduction, Canny Edge Detection is applied. Canny can then identify the actual defect boundaries more accurately.

So the correct sequence is:

Original Image



Median Filtering



Noise Removed



Canny Edge Detection



Important Edges Detected



Defect Identification

### **(e) Present your response in a clear and structured format.**

Input: Industrial inspection image

Step 1 – Median Filter:

Removes unwanted noise while preserving important image structures.

Step 2 – Canny Edge Detection:

Detects the boundaries and edges of cracks, scratches, and other defects.

Step 3 – Defect Analysis:

The detected edges can be used to identify and analyze defects.

Conclusion: The combination of Median Filtering followed by Canny Edge Detection is suitable because the first step removes noise and the second step detects important edges. This provides a cleaner image and improves the accuracy of industrial defect detection.